A-A-54576
24 January 1992

COMMERCIAL ITEM DESCRIPTION

AUDIOMETER

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a microprocessor controlled screening audiometer with integral electrostatic printer.

Salient Characteristics:

General:

1. The audiometer shall comply with all applicable requirements of the following publications, of the issue in effect on the date of solicitation:

    a. ANSI S3.6 - Specifications for Audiometers
    b. UL 544 - Electric Medical and Dental Equipment

2. Electrical requirements:

    a. The audiometer shall be capable of operating over the full tolerance range of one or more of the following line voltages and frequency combinations as specified by the customer requisition:

115VAC plus or minus 10 percent, 50 plus or minus 2Hz.
115VAC plus or minus 10 percent, 60 plus or minus 2Hz.
230VAC plus or minus 10 percent, 50 plus or minus 2Hz.
230VAC plus or minus 10 percent, 60 plus or minus 2Hz.

The dual voltage and hertz feature is desirable but not mandatory.

    b. Change over devices, switches, circuits, or transformers if used for the purpose of voltage or frequency conversion, shall be clearly marked and identified in the

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appropriate documentation which accompanies the unit. Mechanical or electrical adjustments to effect the conversion shall not require the use of specialized tools or skills.

3. Installation shall be by two patch cords only. Audiometers shall be compatible with existing booths, which have only one earphone jack and one subject response jack.

4. A certificate of calibration shall be supplied with each audiometer.

5. All software updates/product improvements shall be made available.

Operational:

1. Test frequencies: 500, 1000, 2000, 3000, 4000 and 6000 hertz (Hz) presented in the automatic mode.

2. Frequency sequence: first ear - 1000, 2000, 3000, 4000, 6000, 1000 (retest), 500; the same sequence shall be followed in the second ear.

3. Ear sequence: Right ear first with manual override to test left ear first.

4. Hearing level intensity range: from 0dB to not less than 100dB, in 5dB steps.

5. Frequency accuracy: Less than one percent error.

6. Amplitude linearity: Within 1dB for every 10dB increment with a maximum accumulated error of 2dB or less relative to the calibration level.

7. A manual mode shall be available for retests. The manual mode shall be programmed to present the identical tone sequence as the automatic mode. The program shall be identical to the automatic mode in number of acceptable responses. The manual mode shall not be open to operator manipulation for acceptance of fewer responses than the automatic mode.

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8. A complete bilateral test with a variable interstimulus interval shall take no longer than six minutes with a fully cooperative, normal hearing individual.

9. Stimulus characteristics:

a. Tone character: 1.2 seconds pulse train. Tone shall be on for 0.2 seconds and off for 0.2 seconds (50 percent duty cycle with three pulsed tone presentations). Rise and fall times shall correspond with values in ANSI S3.6.

b. Shall operate only in the pulsed tone mode.

c. Time between pulse trains shall vary randomly between one and two seconds.

d. Patient response window duration shall be 1.8 seconds. Window shall begin 50 milliseconds after the onset of the pulse train and end not less than 10 milliseconds before the onset of the following pulse train.

e. Pulse train shall terminate when the patient response switch is depressed.

f. Test paradigm shall be the modified Hughson - Westlake procedure as specified in ANSI S3.21-1978, paragraphs 4.1(2) and 4.2. Threshold criteria shall be two responses at the same level after the initial response. Starting level shall be 30-40dB; if no response, increase level 20-30dB.

g. Shall retest at 1000Hz in both ears for plus or minus 5dB agreement with initial threshold.

10. Error warnings: Shall have visual error warnings and auditory signals to alert the technician to patient or test errors. The following error conditions shall be detected and signaled:

a. Threshold not established at 1000Hz, either ear.

b. Failure of retest at 1000Hz. Retest at 1000Hz shall be within plus or minus 5dB of initial test at 1000Hz. If not, test shall abort.

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c. Patient response switch not released.

d. Patient response with no tone present (outside of response window). Error warning shall come after three occurrences of this condition.

e. Invalid frequencies (less than three), manual retest. Frequencies to be retested shall be manually selected, one at a time. Retest shall be performed automatically after selection of individual frequency to be retested.

f. Invalid frequencies (three or more), complete manual retest.

11. Shall be capable of interfacing with a data storage device, such as a modular cassette or micro-floppy disk storage unit, to be used in computerized hearing conservation data registry (i.e. shall have an RS 232C interface).

12. Shall have an LED or LCD digital or alphanumeric display for frequency tested, hearing level, and error codes.

13. Shall have individual talkover (intercom communications) capability, with built-in (internal) microphone.

14. Shall be programmed to compute and print-out Army hearing loss profile codes H1, H2, H1/H3E, H2/H3E, and H3/H3E, as defined in Army Regulation AR 40-501.

15. The audiometer shall provide for operator entry, via a keyboard or touchpad, of data which will appear on the printout. The Social Security number shall be entered by the operator at the start of each examination. The certification number shall be entered when the unit is turned on and shall remain in memory until changed by the operator or the unit is turned off. The current date and time shall be entered when the unit is turned on and shall remain in memory, automatically updated by an internal clock, until changed by the operator or the unit is turned off. The date last calibrated shall be entered by maintenance personnel and shall remain in memory until changed by maintenance personnel. The use of DIP switches or similar devices for the calibration date is acceptable.

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16. The audiometer shall have a built-in electrostatic printer which shall deliver an archive-quality copy that is readable after a 50-year storage. The printout shall be automatically generated at the completion of a test in the automatic mode. It shall also be possible to manually initiate a printout. The unit shall be programmed to print the following information:

* Name _____
* Social Security Number (automatically printed from memory)
* Date and time of examination (automatically printed in MM-DD-YY HH:MM format)
* Technician's signature _____
* Certification number (automatically printed from memory)
* The audiogram, including the frequency in Hz and the dB HTL
* Make, model and serial number of the audiometer
* Calibration standard (for example, ANSI, S3.6-1989)
* The hearing loss code (H1, H2, H1/H3E, H2/H3E, or H3/H3E)
* Date last calibrated (automatically printed in MM-YY format)

The printout may be arranged in the manufacturer's standard format, so long as all the above information is contained on the printout. The actual social security number, certification number, calibration date and date/time of examination shall be printed automatically. Name and technician's signature shall be followed by a blank line to allow signatures to be written after printout is generated.

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17. Shall have TDH-39 earphones with MX-41/AR cushions mounted in appropriate earphone enclosures. The suspension system of the earphones shall allow the cushions to come in contact with the pinnae (external ears) with normal pressure and shall fully enclose the external ears with noise-excluding shells.

Maintenance, repair and calibration:

1. Shall have a calibration servicing switch to allow the automatic setting of the audiometer to a 70dB continuous tone at 500Hz, with subsequent manual selection of frequency and earphone. An alternate method of producing the specified calibration tone, in lieu of a dedicated switch, would be to push a sequence of buttons on the keyboard to generate the required tone. Audiometers which require any or all of the buttons to be continuously held down during this process shall not be acceptable.

2. Signal-to-noise shall be as specified in ANSI S3.6.

3. Electronic circuit boards and connector cables shall be designed so that they cannot be inserted into the wrong circuit or chassis (keyed so that they can only be inserted into the proper circuit or chassis).

4. The electronic circuitry of the audiometer shall be designed to self-diagnose malfunctions to the maximum extent practical.

5. Audiometer shall have a secured access for calibration. The earphone shall be individually calibrated by either activating an EEPROM from the front panel controls or by trim pots within the unit.

6. Operation and service data manuals:

a. Information content:

The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures or illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures

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shall be written in the English language and all schematics and wiring diagrams shall use American electrical and electronics symbols. The manuals shall include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc. pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

b. Quantities, distribution and schedules:

Two complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of two service manuals need to be supplied to each consignee. However, 2 operation manuals still need to be supplied with each unit.

Workmanship. The audiometer shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One audiometer, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one audiometer.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-2. The unit of product for test purposes shall be one audiometer.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

a. Verify the unit operates on each specified voltage and frequency. All remaining tests are to be performed using each specified input power.

b. Verify installation by two patch cords only, one earphone cord and one subject response cord.

c. Verify frequency sequence, in automatic mode, of 1000, 2000, 3000, 4000, 6000, 1000, 500 Hz.

d. Verify actual frequencies generated are within one percent of indicated frequencies.

e. Verify amplitude linearity is within 1dB for every 10dB increment, with maximum accumulated error of 2dB or less relative to calibration level.

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f. Verify time to complete bilateral test on a normal hearing individual is no more than 6 minutes.

g. Verify duration of the pulse train is 1.2 seconds, with rise and fall times as specified in ANSI S3.6.

h. Verify duration of the patient response window is 1.8 seconds.

i. Verify pulse train terminates when response switch is depressed.

j. Verify error warnings are given for error conditions listed under Salient Characteristics, Operational, paragraph 10, a-f.

k. Verify the printout contains all information specified under Salient Characteristics, Operational, paragraph 16.

l. Verify calibration tone is produced by either a dedicated switch or a specific sequence of keyboard buttons, with no button continuously held down during the process.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be

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otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Suppart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.This Medical Procurement Item Description covers a microprocessor controlled screening audiometer with integral electrostatic printer.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

6515-01-305-1161

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project Number: 6515-5170

Location: AUDIOCID.16D

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2 The submitter of this form must complete blocks 4, 5, 6, and 7.

3 The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I. RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54576	2. DOCUMENT DATE (YYMMDD): 24 January 1992

3. DOCUMENT TITLE: AUDIOMETER

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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